

A Hybrid Preprocessor Architecture for the Streaming Spectrum Sensor

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Abstract—Previous work introduced the streaming spectrum sensor (StSS), a low-memory field-programmable gate array architecture that serially evaluates trial symbol-rate and center-frequency pairs. This paper describes a fast Fourier transform (FFT) symbol-rate preprocessor intended to reject most trial cycle-frequency bins before streaming spectral correlation analysis. In an illustrative case, reducing 11 downstream evaluations to two gives a 5.5-fold upper bound on the speedup before preprocessor overhead. A two-detector implementation is reported to fit a Virtex-7 XC7VX485T. Because the preserved results contain neither a serial baseline nor post-preprocessor detection, timing-closure, or power measurements, they establish implementation fit rather than measured end-to-end throughput or edge-device suitability.

Index Terms—streaming spectrum sensor, spectral correlation, RF feature detection, FPGA, edge computing

I. INTRODUCTION

Spectrum sensing can provide environmental knowledge that a cognitive radio uses to adapt its operation. One family of feature detectors estimates a signal’s spectral correlation function (SCF), also called cyclic spectral density. Common estimators include the strip spectral correlation analyzer and FFT accumulation method; their resource and resolution tradeoffs are reviewed in [1].

The streaming spectral correlation analyzer (StSCA) was developed to estimate selected SCF points with low block random access memory (BRAM) use [1]. The streaming spectrum sensor (StSS) adds peak detection and a power-scaled threshold to estimate the symbol rate and center-frequency offset of active signals [2]. Related cognitive-networking applications are discussed in [2], [3]. This paper describes an FFT preprocessor and scheduling updates intended to reduce serial StSCA work. The preserved evidence supports resource-utilization claims for one FPGA configu-

ration; claims that require new measurements are identified explicitly.

II. BACKGROUND

Let $X[n]$ be a complex sample sequence acquired at sample rate f_s , let f_k be a trial frequency shift, and let α_v be a trial cycle frequency. For an N -sample record, the finite-record StSCA estimate can be written as

$$y_{v,k}[n] = \sum_{\ell} h_v[\ell] X[n - \ell] e^{-j2\pi f_k(n-\ell)/f_s}, \quad (1)$$

$$\hat{S}_X^{\alpha_v}(f_k) = \frac{1}{N} \sum_{n=0}^{N-1} |y_{v,k}[n]|^2 e^{-j2\pi \alpha_v n/f_s}. \quad (2)$$

Here, $h_v[\ell]$ is the low-pass filter selected for cycle-frequency branch v . Equation (2) estimates one configured (α_v, f_k) point rather than the continuous SCF; its derivation and mapping to symbol rate and center frequency are given in [1].

The center-frequency detector (CFD) applies the test

$$|\hat{S}_X^{\alpha_v}(f_k)|^2 \underset{H_0}{\overset{H_1}{\geq}} \beta_{\text{CFD}} \left(\frac{1}{N} \sum_{n=0}^{N-1} |X[n]|^2 \right)^2. \quad (3)$$

The dimensionless multiplier β_{CFD} sets an operating point; it is not itself the probability of false alarm. The calibration in [2] was specific to that paper’s waveform, channel, search grid, and decision objective. The reported validation scope is M-ary quadrature-amplitude modulation (M-QAM) with square-root-raised-cosine (SRRC) pulse shaping. Operation on other modulation and pulse-shape families was not demonstrated.

III. DESIGN OVERVIEW

A. Symbol-Rate Preprocessor

The original StSCA avoids FFT cores by processing one configured (α_v, f_k) point in a streaming fashion. The preprocessor concept—an FFT of the instantaneous power followed by candidate selection—was described by Koch and Downey [4] and recommended for the StSCA in [1]. The present architecture integrates that concept with the StSS so

This work was supported by the National Aeronautics and Space Administration’s (NASA) Space Communications and Navigation Program (SCaN). This work was performed by NASA Glenn Research Center’s (GRC) Cognitive Signal Processing Branch to support NASA’s Cognitive Communications Project.

Post-publication fact-checked author revision, September 2026. The version of record appeared in the 2023 IEEE Space Hardware and Radio Conference (SHaRC), pp. 34–36, doi: 10.1109/SHaRC56958.2023.10046267.

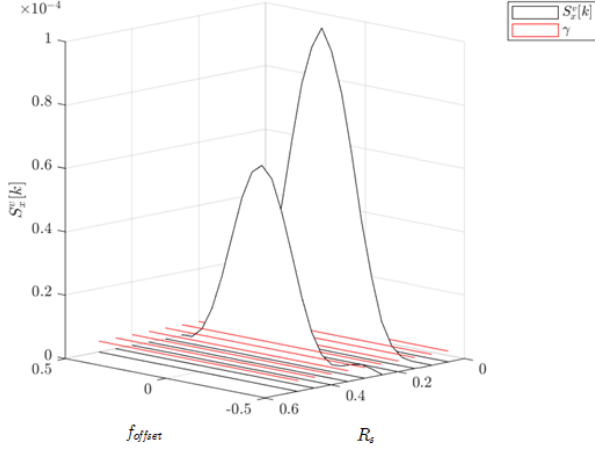


Fig. 1. Illustrative SCF result for two simulated in-band signals: ($R_s = 0.10$ MBd, $f_{\text{offset}} = -0.20$ MHz) and ($R_s = 0.25$ MBd, $f_{\text{offset}} = 0.20$ MHz). The threshold test is applied to the squared-magnitude statistic in (3).

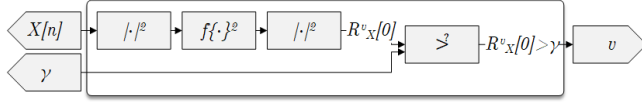


Fig. 2. SRP block diagram. The FFT computes all N DFT bins for the instantaneous-power sequence; the earlier StSCA evaluates configured DFT bins serially.

that only selected cycle-frequency candidates are sent to the CFDs.

Figure 1 illustrates 11 evaluated cycle-frequency bins, of which two exceed the plotted threshold. Skipping nine bins reduces downstream evaluations by 81.8%. If each serial CFD evaluation costs T_{CFD} and the preprocessor costs T_{SRP} , the corresponding ideal work ratio is $11T_{\text{CFD}}/(T_{\text{SRP}} + 2T_{\text{CFD}}) < 5.5$. Thus, 5.5 is an upper bound for this example, not a measured throughput gain.

For the SRP, the cyclic autocorrelation estimate at lag $\tau = 0$ is a DFT of the real-valued instantaneous power:

$$\hat{R}_X^{\alpha_m}[0] = \frac{1}{N} \sum_{n=0}^{N-1} |X[n]|^2 e^{-j2\pi mn/N}, \quad \alpha_m = \frac{mf_s}{N}. \quad (4)$$

Because $|X[n]|^2$ is real, only one spectral half is unique. The saved RTL tests $m = 2, \dots, N/2 - 1$, excluding DC, bin 1 as a bleedover guard, Nyquist, and negative frequencies. It does not group adjacent candidates; leakage and family-wise false alarms therefore require full-grid calibration. SRP and CFD scaling also requires separate calibration.

FFT storage and latency are architecture-dependent; an N -point core requires $O(N)$ state [5]. The saved 65,536-point, 16-bit fixed-point core is pipelined, unscaled, natural-order, and BRAM-backed. Its XCKU5P checkpoint and conflicting 100/125-MHz metadata do not identify the Table I build.

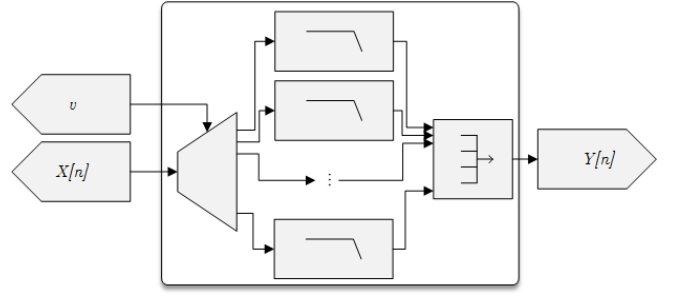


Fig. 3. Selectable low-pass-filter bank. Branch v selects a filter with cutoff $f_{c,v}$; the coefficients, cutoff convention, gain, and equivalent noise bandwidth are required to reproduce the detector calibration.

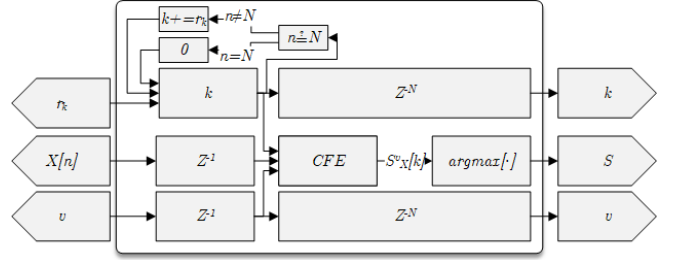


Fig. 4. CFD block diagram. The trial center frequency f_k advances by step r_k ; for a selected α_v , peak selection operates on $|\hat{S}_X^{\alpha_v}(f_k)|^2$.

B. Center-Frequency Estimator

Earlier simulation found a fixed two-tap smoother adequate for its tested cases, not for general convergence [1]. The saved snapshot contains 128 banks of 41 signed 16-bit FIR taps, spaced by 488.28125 kHz at $f_s = 125$ MHz, but no before-and-after metric or noise-bandwidth calibration.

C. Center-Frequency Detector

The CFD retains the functionality in [2], with thresholding moved to the StSS and SRP. In the saved snapshot, sweep wrap resets before comparing the final center-frequency result, omitting that endpoint; output alignment also requires re-verification.

D. Symbol-Rate Distributor

The SRD replaces the earlier polling distributor and FIFO commutator [2] with one-hot priority logic. The saved active path has no candidate backpressure: it drops candidates when all CFDs are busy and results when their FIFOs are full, without retry or loss counters. Resource savings and reliability were not measured.

E. Streaming Spectrum Sensor

The top-level StSS contains the threshold logic, SRP, SRD, and result FIFOs and tracks sample and result counts. It interfaces with an analog-to-digital converter and a CPU-based dynamic spectrum manager (DSM).

Additional CFDs increase concurrent service capacity until another component becomes the bottleneck; they do

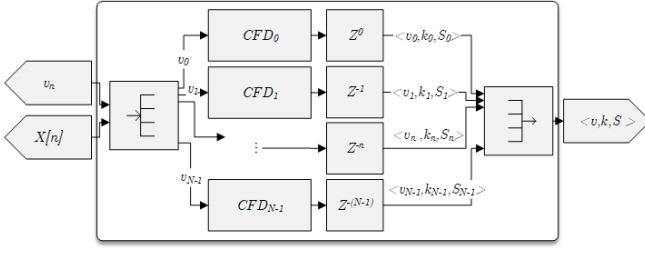


Fig. 5. SRD block diagram. A candidate from the SRP is assigned to a ready CFD; if none is ready, the candidate is discarded. Each CFD starts at the beginning of an N -sample frame; index-based staggering permits result commutation under the assumed fixed-latency, no-stall schedule.

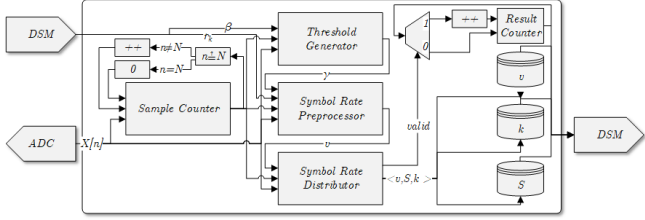


Fig. 6. StSS block diagram. The DSM configures threshold multipliers and center-frequency search resolution. A target false-alarm probability must be mapped to each multiplier by calibration for the selected waveform, channel, grid, filters, and fixed-point scaling.

TABLE I
REPORTED XC7VX485T RESOURCE UTILIZATION

Resource	Used	Available	Utilization
Slice LUTs	22,463	303,600	7.40%
Flip-flops	15,426	607,200	2.54%
36-Kb BRAM equiv.	186.5	1,030	18.11%
DSP48E1 slices	284	2,800	10.14%

not reduce the statistical precision required for threshold calibration.

IV. RESULTS AND LIMITATIONS

The Table I capacities identify the reported target as a Virtex-7 XC7VX485T [6]; $xc7k7-t$ is invalid, and the package, speed grade, and build remain unknown. The reported design used 16-bit data, $N = 2^{16}$, and two CFDs. The saved configuration reconstructs 34.36 s as a full-resolution sweep: for $FcStep = 1$, $f_s = 125$ MHz, one

sample per clock, and no stalls, $2^{16}N/f_s = 34.359738368$ s. Two CFDs parallelize candidates, not an individual sweep. This is a conditional calculation, not a retained measurement or throughput comparison.

The four percentages are arithmetically consistent with the reported counts. The fractional BRAM value denotes an 18-Kb block counted as half of a 36-Kb equivalent. One utilization point demonstrates fit on this device but cannot establish scaling with CFD count, timing closure, power, reliability, or suitability for smaller edge FPGAs.

V. CONCLUSION

The architecture combines an instantaneous-power FFT preprocessor with the StSCA and a two-CFD distributor. The reported counts fit an XC7VX485T, but the saved artifacts do not identify their build or demonstrate throughput or calibrated detection. Reproduction requires an identified source revision and FPGA part, a portable self-checking testbench, configuration and synthesis reports, baseline comparison, drop and P_d/P_{fa} measurements, and a CFD-count sweep. Goertzel DFTs and normalized coherence remain proposals to benchmark.

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